

L2 - List & Tuples

- A list is an ordered collection of objects. It can store values of the same or different data types.
- We can create list. $n = [, , , ,]$
- We can access elements of list. `print(n[0])`
- Lists are mutable. Their contents can be changed after creation.

`n[1] = 10` or `n.append(12)`

- An empty list has no elements. Before we can assign to the i^{th} position, that posⁿ must exist. We start with empty list & use `append()` to add one element at a time. This allows the list to grow dynamically.
- Lists can be combined using `+` operator. It creates a new list.
- Add several elements using `extend()`. It modifies the original list itself.
- We can remove an element by index.
`del(l[2])`
- We can remove an element by its value.
`l.remove(25)`
 ↳ remove its first occurrence
- `l.pop(i)` → remove & return element at index i .
- `l.pop()` → remove & return last element (-1) .
- A string can be split into a list using `split()`.

```
text = "a b c"
print(text.split())
⇒ ["a", "b", "c"]
```

- A list of strings can be joined using `join()`
`words = ["a", "b", "c"]`
`text = " ".join(words)`

- some useful list operations - `len(l)`, `sum(l)`, `min(l)`, `max(l)`, `l.sort()`, `l.reverse()`, `7 in l`, `3 not in l`.
- methods like `sort`, `reverse`, `extend`, `append`, `remove` & `pop` modify the list.
- `M = L` $\Rightarrow$ does not create a new list.
 $\hookrightarrow$ both refers to the same object.
- In python, variable is a name that refers to an object. This avoids unnecessary copying of large objects.
- If you want a separate copy, `M = L[:]`
Also, `M = L.copy()`, `M = list(L)`
- A tuple is an ordered collection of obj.
Create tuples $\Rightarrow p = (3, 4)$
elements can be accessed $\rightarrow p[0]$
Tuples can't be modified, once made.
- A tuple is useful when values naturally belong together and should not change.
Attempting to modify a tuple produces an error.
- A function sometimes need to return more than 1 value. A tuple provides a convenient way to group these values together.

```
def _____ :  
    return (a1, a2)  
           tuple
```